

# VENKATRAMAN R

Data Science & AI Engineer | ML & Analytics & GenAI

+91 97917 93938    ✉ venkatraman0400@gmail.com    ✉ venkatraman0400@gmail.com    🌐 github.com/venkatraman0400-blip

## PROFESSIONAL SUMMARY

Aspiring Data Science & AI Engineer with a hands-on portfolio spanning fraud detection, anomaly detection, automated ETL pipelines, and GenAI applications. Proficient in Python, XGBoost, SMOTE, SHAP, LangChain, and OpenAI API. Passionate about building ML solutions that deliver measurable business impact. Seeking entry-level roles in Data Science, Machine Learning, and Generative AI.

## TECHNICAL SKILLS

|                          |                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Languages</b>         | Python, SQL                                                                                                                           |
| <b>ML / AI</b>           | Scikit-learn, XGBoost, Random Forest, Logistic Regression, SMOTE, Anomaly Detection, SHAP, Pandas, NumPy, Matplotlib, Seaborn, Plotly |
| <b>GenAI / LLM</b>       | LangChain, OpenAI API, HuggingFace Transformers, Prompt Engineering, RAG, Streamlit                                                   |
| <b>BI &amp; Scraping</b> | Power BI, Tableau, Excel (Advanced), Power Query, BeautifulSoup, Selenium                                                             |
| <b>Dev Tools</b>         | Git, GitHub, Jupyter Notebook, VS Code, Google Colab                                                                                  |
| <b>Competencies</b>      | Feature Engineering, EDA, Model Evaluation (AUC/ROC/F1), Imbalanced Data, ETL Pipelines, Data Wrangling                               |

## PROJECTS

### FraudGuard Mobile Transaction Fraud Detection Completed

Python | XGBoost | Scikit-learn | SMOTE | Pandas | Seaborn

- Built end-to-end fraud detection pipeline on imbalanced data; applied SMOTE achieving **AUC 0.97+** and **F1-score 0.92** on the fraud class.
- Engineered transaction-velocity, amount-deviation, and time-of-day features; delivered ROC curve and feature importance charts to communicate risk drivers.

### BankShield Bank Account Fraud Detection Completed

Python | XGBoost | Random Forest | Scikit-learn | Pandas | Matplotlib

- Built supervised fraud classifier with full ML pipeline EDA, feature engineering, cross-validation achieving **precision 0.94** and **recall 0.89**.
- Applied threshold tuning to optimise precision-recall trade-off; produced SHAP-based explainability report identifying top fraud risk drivers for stakeholders.

### DataHunt Automated Web Scraping & ETL Pipeline Completed

Python | BeautifulSoup | Selenium | Pandas | Power BI

- Automated extraction of 1,000+ structured records from e-commerce and job portals; cleaning workflows cut manual collection time by **~70%**.

### SalesPulse Interactive Business Dashboard Completed

Excel | Power Query | Pivot Tables | Slicers

- Built dynamic multi-KPI dashboard tracking revenue trends and regional sales; automated data refresh via Power Query, reducing manual reporting time by **60%**.

### AskMyData AI Q&A App with LLMs In Progress

Python | LangChain | OpenAI API | RAG | Streamlit

- Building GenAI app where users upload datasets and query in plain English; LangChain agents auto-generate Pandas/SQL analysis code using RAG and prompt engineering.

## EDUCATION

### Professional Certification in Data Science with Artificial Intelligence Jun 2026

Boston Institute of Analytics

### B.E. Automobile Engineering | CGPA: 7.25 Sep 2021 May 2024

VISTAS (Vels Institute of Science, Technology & Advanced Studies)

### Diploma in Automobile Engineering | 72% Jun 2016 Jun 2019

Hindustan Institute of Engineering Technology

## VOLUNTEERING

Blood Donation @ Hindu Mission Hospital (Mar 2023) & Sri Ramachandra Hospital (Jul 2024)